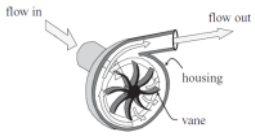
Week of 7th Nov.

Tuesday, 7 November 2023 9:43 AM

Pump Characteristics Curves

Pumps → mechanical devices that allow for fluid transportation via pipes (Section 9.2.4.1)

↳ Centrifugal pumps are a common type → transfer rotational energy (from electric motor) to fluid movement



+ Working principle is energy balance! Electric energy → shaft work (by pump to fluid) → affects speed, pressure & other fluid characteristics

+ Pump manufacturers need to characterize a pump → pump operation under various conditions [Pump curves!]

$\Delta E = Q + W$ (Work done on system by environment)

ΔE: total energy of system; Q: Heat from environment to the system via boundary

Energy balance on CV: $\frac{dE_{cv}}{dt} + \iint_{CS} (\hat{n}\cdot\underline{v})\rho\hat{E}\,dS = \dot{Q}_{in,cv} - \dot{W}_{by,cv}$ $\quad [E_{cv} = U + E_K + E_P]$

Closed system

Open system (mass flow in & out)

Some assumptions

$\Delta E_p + \Delta E_K + \Delta U = Q_{in,cv} - W_{by,cv}$ $\quad (W_f = F\cdot d = (P\cdot A)\cdot d)$

$W_{by,cv}$: W_s Shaft work (work performed by fluid due to moving parts); Flow work (W_f) (work done by fluid as it enters & leaves the system)

$H = U + PV$

SS Macro Energy balance

$$\Delta E_p + \Delta E_K + \Delta H = Q_{in,cv} - W_{s,by\,cv}$$

Single in, single out, Incompressible fluid

$$\dot{m}\left(\frac{\langle v\rangle_2^2}{2} - \frac{\langle v\rangle_1^2}{2}\right) + \dot{m}g(z_2 - z_1) + \dot{m}\left(\hat{U}_2 - \hat{U}_1 + \frac{P_2}{\rho} - \frac{P_1}{\rho}\right) = \dot{Q}_{in,cv} - \dot{W}_{s,by\,cv}$$

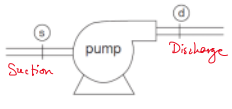
Temp constant, no phase changes/chemical reaction, negligible heat transfer

Mechanical energy Balance (MEB)

$$\frac{\Delta P}{\rho} + \frac{\Delta\langle v\rangle^2}{2\alpha} + g\Delta z + F = -\frac{W_{s,by}}{m}$$

pressure; density; average velocity; $\alpha = 1$ (turbulent flow); friction term $[\Delta U - Q]$

Pump can add energy via:
+ Increase velocity of fluid
+ create pressure differential
+ Pump can change fluid height (raise its potential energy)



η = Pump efficiency $= \frac{\text{(useful energy)}}{\text{(total energy)}}$ (energy converted to work (rather than friction))

$$= \frac{\left(-\frac{W_{s,by}}{m} - F\right)}{\left(-\frac{W_{s,by}}{m}\right)}$$

$$\frac{\left(\frac{\Delta p}{\rho} + \frac{\Delta \langle v \rangle^2}{2\alpha} + g\Delta z\right)}{\left(-\frac{W_{s,by}}{m}\right)}$$

Pump brake horsepower: $bhp = -W_{s,by}$ (Total work delivered by pump)

↓ Dividing by g

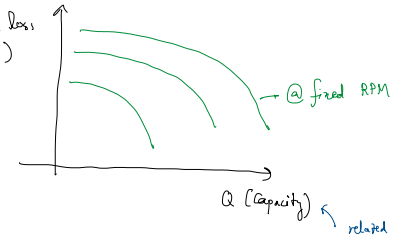
$$\therefore \quad \frac{\Delta p}{\rho g} + \frac{\Delta \langle v \rangle^2}{2\alpha g} + \Delta z = \eta\left(-\frac{W_{s,by}}{mg}\right) = \eta\,(bhp)$$

$$\boxed{\left(\frac{P_d}{\rho g} + \frac{\langle v \rangle_d^2}{2g} + z_d\right) - \left(\frac{P_s}{\rho g} + \frac{\langle v \rangle_s^2}{2g} + z_s\right) = \eta\left(-\frac{W_{s,by}}{mg}\right)}$$

Discharge head (H_d) [=] length — Suction head (H_s) [=] length

$$\underbrace{(H_d - H_s)}_{\text{Head loss}} = \eta\left(-\frac{W_{s,by}}{mg}\right) = \eta\left(-\frac{bhp}{mg}\right)$$

give info about η & bhp



Pumping Ability of a CP:
+ Shaft rotational speed (RPM)
↑ Shape of vanes
+ flow path

Lab: Characterize relationships b/w discharge flow rate, net head, bhp, efficiency in a CP

↳ Vary Q (via rotameter)
↳ Measure pressure loss → calculate head loss (H_{loss} or $H_{net} = H_d - H_s$)
↳ Non-dimensionalize variables & then plot them!

PROBLEM: 8.30 Calculate the extra stress tensor $\underline{\underline{\tau}}$ for potential flow around a sphere of an inviscid, Newtonian fluid. Calculate the total stress tensor $\underline{\underline{\Pi}}$. Comment on what is obtained.

PROBLEM: 8.40 What is the force (i.e., drag) on a thin plate given the following conditions? The fluid is water, the plate is 0.52 m long and 6.3 m wide, and the free-stream velocity is 1.3×10^{-2} m/s.

Boundary layer flow:

$\nabla \cdot \underline{v} = 0$

Navier Stokes

} $\underline{v}$ varying w/ two dimensions → Assumptions → Velocity profile for laminar flow past a flat plate

$$\frac{v_1}{v_\infty} = f'(\zeta) = 1 - (0.5434)\log\left[1 + (68.3)10^{-0.6247\zeta}\right]$$

$$\underline{F} = \iint \left[\hat{n} \cdot \underline{\underline{\pi}}\right]_{@surface} dS$$

$$\tau_w = 0.332 \mu v_\infty \left(\frac{\rho v_\infty}{\mu x_1}\right)^{1/2} \longleftarrow \text{Drag is the force component in flow direction}$$

